Electrical Engineering and Automation Curriculum

(Updated in May 2023)

1. Program Educational Objectives

Electrical engineering and Automation is a multifaceted discipline that over the last century has produced an astounding progression of technological innovations that have shaped virtually every aspect of modern life. Electrical engineers need a broad and solid foundation in mathematics and physics to support their education in the engineering principles of analysis, synthesis, design, implementation, and testing of the devices and systems that provide the bedrock of modern energy, communication, sensing, computing, medical, security, and defense infrastructures. Within each sub-discipline one can find application domains that strongly rely on hands-on experimental work or that are based on theoretical, mathematical, and computational approaches. The multidisciplinary nature of electrical engineering education addresses the growing demand for the innovation and design of sensing, communication, computing, and decision-making systems of increasing complexity in consumer, defense, and medical applications.

2. Graduation Requirement

1) Curriculum Requirement

The curriculum leading to the degree of Bachelor of Science in Electrical Engineering from ZJU requires **135** credits and is organized into required courses; technical electives; liberal education; and other electives.

3. Required Courses

3.1 Orientation and Professional Development

These courses introduce the opportunities and resources our institute and curriculum can offer you as you work to achieve your career goals. They also provide the skills to work effectively and successfully in the engineering profession.

Course Code	Course Name	Credit
ENG 100	Engineering Orientation	1

3.2 Foundational Mathematics and Science

These courses stress the basic mathematical and scientific principles upon which the engineering discipline is based.

Course Code	Course Name	2016-2020	2021-
MATH 221	Calculus Ⅰ	4	4
MATH 231	Calculus Ⅱ	3	3
MATH 241	Calculus Ⅲ	4	4
MATH 257	Linear Algebra with Computational Application		3
MATH 285	Introduction to Differential Equation		3
MATH 286	Intro to Differential Equations Plus	4	
PHYS 211	University Physics: Mechanics	4	4
PHYS 212	University Physics: Elec & Mag	4	4
PHYS 213	University Physics: Thermal Physics	2	2
	Total required	25	27

3.3 Technical Core

These courses stress fundamental concepts and basic laboratory techniques that comprise the common intellectual understanding of electrical engineering.

Course Code	Course Name	2016-
ECE 110	Intro to Electronics	3
ECE 120	Intro to Computing	4
ECE 220	Computer Systems & Programming	4
ECE 210	Analog Signal Processing	4
ECE 313	Probability with Engineering Application	3
ECE 329	Fields and Waves Ⅰ	3

ECE 385	Digital Systems Laboratory	3
ECE 340	Semiconductor Electronics	3
ECE 498FTPT*	Final Term Project/Thesis of B.E.	10
	Total required	37

**Instructed at Yuquan campus*

4. Elective Courses

4.1 Technical Electives

This elective requirement gives each student the freedom to develop a technical course of study in control science and technology of considerable breadth and focus. Students must complete a minimum of total required hours on technical electives, which include at least:

			2016-2020	2021-
1)	20 credits	ECE courses, see below section 4.1.1 for detail.		
2)	2 courses	ECE Labs, see below section 4.1.2 for detail.		
3)		Non-ECE courses, see below section 4.1.3 for detail.	20	20
		Total required	28	26

4.1.1 ECE courses

Students must take at least 20 credits ECE courses from the list below:

Course Code	Course Name	Credits
ECE 310	Digital Signal Processing	3
ECE 330	Power Circuits & Electromechanics	3
ECE 342	Electronic Circuits	3
ECE 350	Fields and Waves II	3
ECE 374	Introduction to Algorithms & Models of Computation	4
ECE 416	Biosensors	3
ECE 441	Physics & Modeling Semiconductor Device	3
ECE 448	Artificial Intelligence	3 or 4
ECE 449	Machine Learning	3
ECE 464	Power Electronics	3

ECE 469	Power Electronics Laboratory	2
ECE498PSOC*	Power System Operation & Control	2
ECE 498FEM*	From Electricity to Movement	2
ECE 498HF*	HVDC Fundamentals	2

**Instructed at Yuquan campus*

4.1.2 ECE Labs

Students must take 2 labs.

1) Hardware Labs

Course Code	Course Name	Credits
ECE 343	Electronic Circuits Laboratory	1
ECE 470	Introduction to Robotics	4
ECE 486	Control Systems	4

2) Software Labs

Course Code	Course Name	Credits
ECE 311	Digital Signal Processing Lab	1
ECE 314	Probability in Engineering Lab	1
ECE 365	Data Science and Engineering	3

4.1.3 Non-ECE Courses

Students are encouraged to build up their cross-discipline study by taking non-ECE technical courses from Civil Engineering, Mechanical Engineering, Computer Science, and others at ZJUI.

Course Code	Course Name	2016-2020	2021-
CEE 310	Transportation Engineering	3	3
CEE 330	Environmental Engineering	3	3
CS 101	Introduction to Computing: Engineering & Science	3	3
CS 225	Data Structures	4	4
CS 412	Introduction to Data Mining	3 or 4	3 or 4
CS 450	Numerical Analysis	3 or 4	3 or 4
CHEM 102	General Chemistry Ⅰ	3	3

CHEM 103	General Chemistry Lab Ⅰ	1	1
CHEM 104	General Chemistry II	3	3
CHEM 105	General Chemistry Lab II	1	1
MATH 213	Introduction to Discrete Mathematics	3	3
PHYS 214	University Physics: Quant um Physics	2	2
RHET 101	Principles of Writing	3	
RHET 102	Principles of Research	3	

4.2 Liberal Education

The liberal education courses develop students' understanding of human culture and society, build skills of inquiry and critical thinking, and lay a foundation for civic engagement and lifelong learning. To satisfy the liberal education requirements, students must take all courses for grade and complete courses based on the table below:

Category	Credit
1) Humanities & Arts	6
Two courses	
2) Social & Behavioral Sciences	6
Two courses	
3) Culture Studies	6
Western/Comparative Culture(s)	
One course	
Non-Western Culture(s)	
One course	
Total required	18

Appropriate choices in the Social and Behavioral Sciences and in the Humanities and Arts will ensure that these courses also fulfil the requirements in the areas of Western and Non-Western Cultures.

4.3 Free Electives

These unrestricted electives give the student the opportunity to explore any intellectual area of unique interest. This freedom plays a critical role in helping students to define their research specialties. Students are encouraged to take cross-disciplinary courses as free electives.

Free electives	2020-2021	2023-
Total	6	5

5. Chinese Language and Society

These courses are compulsory for all international students. They introduce modern Chinese history, social development, government policies, etc., and help students improve their Chinese and understanding about China.

International students need to complete Chinese language study and Chinese society to fulfill the graduate requirement.

Course Code	Course Name	2016-2022	2023-
CHIN 1001	Chinese Ⅰ	4	4
CHIN 1002	Chinese Ⅱ	5	4
CHIN 1003	Chinese Ⅲ	4	4
CHIN 1004	Chinese Ⅳ	4	4
CHIN 1005	International Chinese Language Proficiency and Testing		2
CULT 2001	China Survey	3	3
	Total	20	21